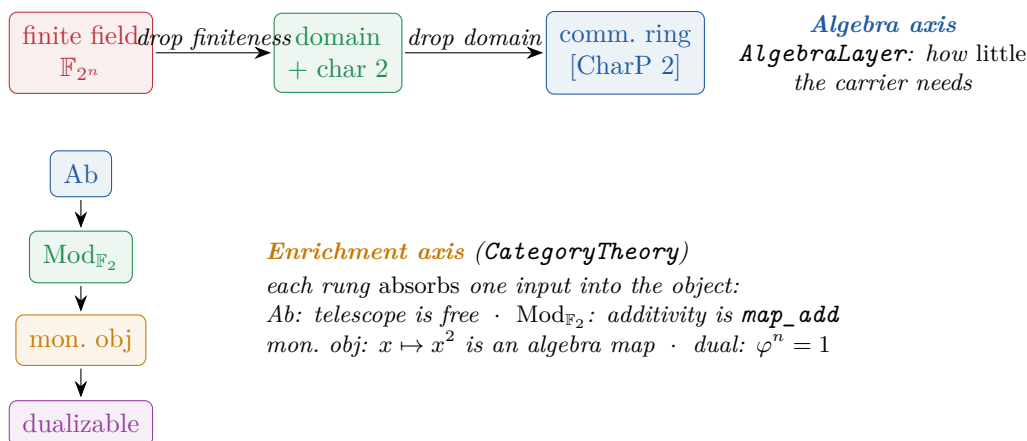


The Algebra and Category Layers of the Dobbertin (1) $\Rightarrow$ (2) Formalisation

A visual field-guide to AlgebraLayer, CategoryTheory, Categorical, ObjectFirst

One picture. The whole first step of Dobbertin’s Theorem 1 is one endomorphism φ (Frobenius, $x \mapsto x^2$) of one object, and one identity, the telescope $(\varphi - 1) \cdot \sum \varphi^i = \varphi^\ell - 1$. This note draws *where each hypothesis lives*: the algebra layer strips the carrier down to a bare commutative ring of characteristic 2 and isolates the *single* spot that needs a field; the category layer builds φ and the telescope up an *enrichment ladder* in which every rung deletes one input by absorbing it into the ambient object.

1 The two axes



module	what it shows	headline
AlgebraLayer	the trace/telescope layer over <i>any</i> comm. ring of char 2; field used once	alg_equation2_of_equation1
CategoryTheory	φ and telescope up the enrichment ladder	equation2_of_equation1_cat
Categorical	the four abstract patterns behind the bricks	(structure, not proof)
ObjectFirst	the type-class bundle collapsed into the object	equation2_of_equation1_obj

2 Algebra layer: how little the carrier needs

The entire telescope layer is proved with `[CommRing R] [CharP R 2]` only. Field-ness enters at exactly *one* arrow — the division by x^{2^k} .

```

proved with only [CommRing R] [CharP R 2]
frobA, frobAEndo (Frobenius; additivity = add_pow_char_pow)
loopA = iterSum; traceA, partialTraceA, numeratorSumA
loopA_telescope (Artin-Schreier, from iterSum_telescope)
traceA_frob_fixed:  $\text{Tr}(x)^{2^k} = \text{Tr}(x)$  (from frobAEndo  $1^n = 1$ )
partialTraceA_telescope:  $S(x) + P(x) = x^2 + x$ 

```

alg_linearized_mul: $x^{2^k} \cdot \ell(x) = 0$ (no division)

↓ divide by x^{2^k}

alg_linearized_eq_zero (the only [NoZeroDivisors R]):
cancel $x^{2^k} \neq 0 \Rightarrow \ell(x) = 0$

alg_equation2_of_equation1 (headline over R)

```

variable {R : Type*} [CommRing R] [CharP R 2] --  $\mathbb{F}_2$ -algebra, nothing more

lemma alg_linearized_mul (k : ℕ) {c x ∈ S P : R} -- NO field, NO division
  (hS : S = P ^ (2 ^ k)) (hSP : S + P = x ^ 2 + x)
  (hεfix : ε ^ (2 ^ k) = ε) (hsol : S + ε = c * x ^ (2 ^ k + 1)) :
  x ^ (2 ^ k) * (c ^ (2 ^ k) * x ^ (2 ^ (2 * k)) + x ^ (2 ^ k) + c * x + 1) = 0

lemma alg_linearized_eq_zero [NoZeroDivisors R] ... (hx : x ≠ 0) : -- ← field enters here, once
  c ^ (2 ^ k) * x ^ (2 ^ (2 * k)) + x ^ (2 ^ k) + c * x + 1 = 0

```

Why no field above the red line. $\text{Tr}(x)^{2^k} = \text{Tr}(x)$ is traceA_frob_fixed, a corollary of the telescope with $\varphi^n = 1$; it supplies $\varepsilon^{2^k} = \varepsilon$ for $\varepsilon = \alpha \text{Tr}(x)$ *without* needing $\text{Tr}(x) \in \{0, 1\}$. So the field-only fact “the trace is a bit” is never used in the linearization — the only field step is the final cancellation.

The field is one instance

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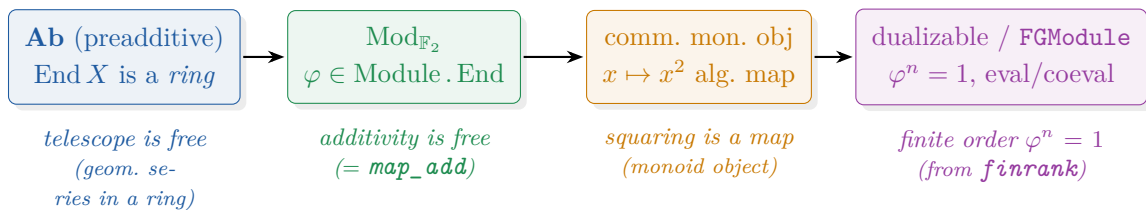
alg_equation2_of_equation1 [CommRing R] [CharP R 2] [NoZeroDivisors R]
  set  $R = \mathbb{F}_{2^n}$  ↓
equation2_of_equation1_of_finField  $\mathbb{F}_{2^n} : \text{Fermat} = \text{frobAEndo\_pow\_card\_field} \Rightarrow \varphi^n = 1$ 

```

The finite field is recovered as the IsDomain specialization: Fermat discharges $\varphi^n = 1$ and field-ness discharges the one division.

3 Category layer: the enrichment ladder

Read the carrier not as a set but as an *object* of a progressively richer category. Each rung to the right makes one hand-proved input *free*.



Rung 1 — Ab. In the endomorphism *ring* of any object of a preadditive category, the telescope is the geometric series — no field, no characteristic, no finiteness:

```

theorem preadditive_telescope (φ : End X) (len : ℕ) :
  (φ - 1) * (∑ i ∈ range len, φ ^ i) = φ ^ len - 1 := mul_geom_sum φ len

```

Rung 2 — $\text{Mod}_{\mathbb{F}_2}$, and the ring isomorphism. Over the base object $\mathbb{F}_2 = \text{ZMod } 2$ the Frobenius is a *morphism of modules* $\varphi = \text{frobLin}$, so its additivity is `map_add`. The ring it lives in is the object's categorical `End`:

$$\begin{array}{ccc} \text{End}(\text{ModuleCat.of } \mathbb{F}_2 F) & \xrightarrow[\simeq_{+*}]{\text{endRingEquiv}} & \text{Module.End } \mathbb{F}_2 F \\ \text{frobMor} \longmapsto & & \text{frobLin} \end{array}$$

so `frobMor_telescope` (run as arrow data) transports to `frobLin_telescope`.

Rung 3 — commutative monoid object. A commutative monoid object of $(\text{ModuleCat } \mathbb{F}_2, \otimes)$ is a $\mathbb{F}_2$ -algebra (`MonModuleEquivalenceAlgebra`); this is *why* $x \mapsto x^2$ is an algebra map, hence why `frobLin` exists at all.

Rung 4 — dualizable / finite. A finite field is a finite-dimensional, hence *dualizable*, object of $\text{FModuleCat } \mathbb{F}_2$. Its finite *dimension* — not raw cardinality — forces $\varphi^n = 1$. The annihilating polynomial $X^n - 1$ is read off through a *monoid map*:

$$\begin{array}{ccc} (F \rightarrow_a [\mathbb{F}_2] F) & \xrightarrow[\rightarrow*]{\text{algEndToLin}} & \text{Module.End } \mathbb{F}_2 F \\ \text{ord=finrank} \downarrow & & \downarrow (-)^n \\ \text{frobLinAlgHom} & \longmapsto & \text{frobLin}, \quad \text{frobLin}^n = 1 \\ \text{ord}=[F:\mathbb{F}_2] \curvearrowright & & \end{array}$$

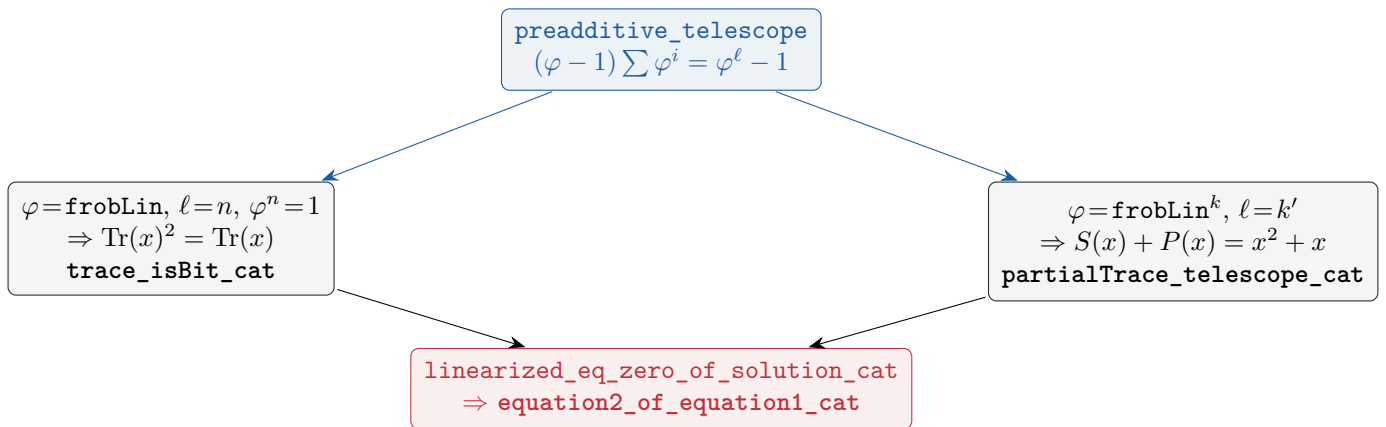
```

lemma frobLin_orderOf : orderOf (frobLinAlgHom (ZMod 2) F) = Module.finrank (ZMod 2) F
lemma frobLin_pow_finrank : (frobLin (F := F)) ^ (Module.finrank (ZMod 2) F) = 1
  := by rw [← algEndToLin_frobLinAlgHom, ← map_pow, ← frobLin_orderOf,
    pow_orderOf_eq_one, map_one] -- φ^n = 1 from finite dimension

```

4 Reassembly: the telescope drives the two facts

Both facts the paper uses are the *one* telescope, applied to φ (at length n) and to φ^k (at length k').



5 Object-first: collapse the type-class bundle

The concrete proof carries `[Field F]` `[Fintype F]` `[CharP F 2]` as three inputs. Reading F as an *object* of $\text{Mod}_{\mathbb{F}_2}$ makes `char 2` a *consequence*: the structure map $\mathbb{F}_2 \hookrightarrow F$ is injective.

$$[\text{Algebra } (\mathbb{Z}\text{Mod } 2) \ F] \xrightarrow{\text{algebraMap inj}} \mathbb{F}_2 \hookrightarrow F \xrightarrow{\text{charP_of_inj_algMap}} [\text{CharP } F \ 2] \quad (\text{derived instance objCharP})$$

```

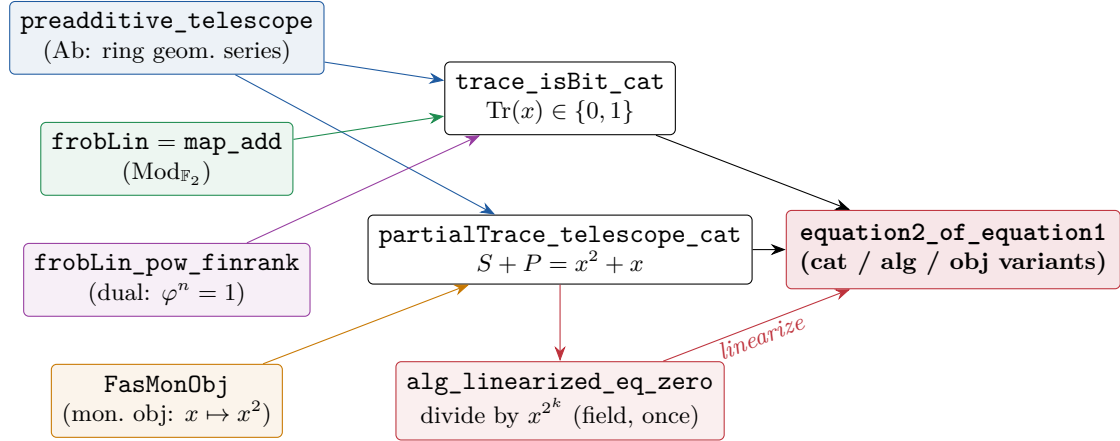
variable {F : Type} [Field F] [Fintype F] [Algebra (ZMod 2) F] -- the object
instance objCharP : CharP F 2 :=                               -- no longer a hypothesis
  charP_of_injective_algebraMap (algebraMap (ZMod 2) F).injective 2

theorem equation2_of_equation1_obj ... -- headline with the collapsed bundle

```

Multiplication itself comes from the monoid object (`objMonObj`, `objAlg`), and the whole categorical chain (`objFrobLin_telescope`, `objFrobLin_pow_finrank`, `objTrace_isBit`) runs uniformly in universe 0.

6 The whole picture



Soundness. `AlgebraLayer`, `CategoryTheory`, `Categorical` and `ObjectFirst` build sorry-free; the headlines (`alg_equation2_of_equation1`, `equation2_of_equation1_cat`, `equation2_of_equation1_obj`, `equation2_of_equation1_of_finField`) depend only on `propext`, `Classical.choice`, `Quot.sound`. The category layer supplies the *shape*; the algebra layer pins the *one* place a field is needed.